

Alexander Fedorako

SOFTWARE ENGINEER

Education

Computer Science and Software Engineering

Language proficiency

English — B2

Domains

HealthCare,
IoT

Software Engineer with 7 years of experience.

I am a results-driven software engineer with a strong background in developing end-to-end solutions across HealthCare and IoT domains. With expertise in Python, I design and implement scalable APIs and microservices tailored to real-world business needs. My experience in the IoT domain includes building robust data pipelines and event-driven architectures that handle real-time telemetry from distributed devices. Additionally, I'm skilled in containerization with Docker, orchestration using Kubernetes, and infrastructure as code with Terraform. Whether optimizing backend performance, designing interactive web interfaces, or building scalable IoT systems, I bring a practical and quality-focused approach to software engineering.

Programming languages

Python.

Backend

Django, Django Rest Framework, FastAPI, OAuth2, Celery, SQLAlchemy, Pydantic, Swagger, Alembic, Pytest, Donesafe API.

Protocols

MQTT, HTTP, BLE.

BI

Sigma BI.

Message brokers

RabbitMQ, Kafka, AWS SQS, Azure Service Bus.

Databases

PostgreSQL, Redis, Cosmos DB, DynamoDB, Timestream.

Natural Language Processing

Hugging Face, LangChain.

Machine Learning

Scikit-learn, Tensorflow.

Cloud

AWS(S3, SES, EKS, ECR, ECS, Lambda, DynamoDB, API Gateway, SageMaker, SNS, SQS, Cognito, RDS, CloudWatch, IAM, etc.),

Azure(Functions, Container Apps, Entra ID, Cosmos DB, Blob Storage, Service Bus, Key Vault, Data Factory, SQL, App Service, etc.).

DevOps

Docker, Docker Compose, Kubernetes (k8s), Jenkins, GitLab CI/CD, Terraform, Nginx, Grafana, Prometheus, Sentry.

Source control systems

Git, GitLab, GitHub.

Projects

**CANCER DETECTION
MEDICAL ASSISTANT**

Project roles

Software Engineer

The project is at the forefront of medical innovation, specializing in the development of state-of-the-art software that harnesses the power of artificial intelligence, particularly through advanced machine learning and image processing techniques.

Period

12.2022 - till now

Responsibilities

- Design microservices architecture and database;
- Backend development;
- Implemented a scalable architecture, transitioning from ECS to EKS for improved performance and reliability;
- Developed and maintained backend services for AI-based oncology detection from MRI scans;
- Preprocessed and cleaned large volumes of medical imaging data for ingestion into AI pipelines;
- Integrated large language models (LLMs) to generate user-friendly medical conclusions and recommendations;
- Setting up monitoring systems to track the performance of deployed ML models;
- Built custom AI pipelines using LangChain to support complex, data-aware workflows;
- Store and manage unstructured data, such as blobs, files, and queues, using AWS S3;
- Using AWS SNS to receive notifications when metric threshold breached in AWS CloudWatch;
- AWS RDS database performance monitoring;
- Building AWS infrastructure with Terraform;
- Automate the deployment process of AWS EKS cluster using Terraform;
- Utilized Celery to process AWS SNS notifications;
- Connect AWS Lambda to events in AWS Cognito;
- Ensured secure user authentication and authorization via AWS Cognito and OAuth2;
- Set up and managed Kafka replication between clusters;

BOAT MONITORING SYSTEM

Boat Monitoring System is a web-based platform designed to collect, aggregate, and visualize data from boats navigating the oceans. The system enables users to monitor real-time boat data, manage fleets, and make data-driven decisions.

- Kafka configuration for communication between services;
- Create and implement SQL database structures to meet application requirements;
- Analyze SQL query performance using query execution plans and optimize as needed;
- Designing and configuring CI/CD pipelines using Jenkins;
- Following TDD principle;
- Mentoring newcomers;
- Code review.

Environment

Python, Django, Django Rest Framework, FastAPI, SQLAlchemy, Pydantic, Alembic, Celery, AWS(S3, SES, EKS, ECR, ECS, Lambda, SageMaker, DynamoDB, API Gateway, SNS, SQS, Cognito, CloudWatch, IAM, Timestream, etc.), Hugging Face, LangChain, Redis, Kafka, PostgreSQL, Jenkins, OAuth2, Docker, Docker Compose, Kubernetes(k8s), GitLab, Terraform, Pytest.

Project roles

Software Engineer

Period

10.2020 - 12.2022

Responsibilities

- Design REST API schemas;
- Designed and implemented ETL pipelines to ingest and transform telemetry data from boats sensors;
- Backend development;
- Build serverless part of the application;

- Preprocessed sensor and navigation data to feed into AI anomaly detection models;
- Set up and configured BI tools to visualize key performance indicators and anomaly metrics;
- Set up AWS Lambda for historical reports generation;
- Implemented a scalable data lake architecture on AWS S3, supporting both structured and unstructured data ingestion;
- Set up Prometheus alerting rules for specific system events and integrated them with Grafana to trigger real-time notifications and dashboards;
- Identify and resolve performance issues with SQL queries, including high resource usage;
- Ensure secure communication between IoT devices and the cloud;
- Implement Kubernetes (k8s) load balancing;
- Develop complex integration tests;
- Streamline and automate the processes for building, testing, and deploying applications;
- Configure GitLab CI for distributed build execution and load balancing;
- Providing clear documentation to API consumers on how to properly integrate and use API keys;
- Review and improve the performance of stored procedures and raw SQL queries;
- Split Celery tasks by priority;
- Configured Prometheus to log and collect metrics from applications and service;
- Integrate Grafana for comprehensive monitoring;
- Implement notifications using AWS SQS;
- Create complex migrations in raw SQL.

Environment

Python, Django, Django Rest Framework, FastAPI,

CLEANING MANAGEMENT SYSTEM

The system is designed for monitoring and managing robots tasked with cleaning commercial and non-commercial facilities. It collects data from robots and static IoT sensors for real-time analysis and storage, providing insights through BI tools. The platform integrates with external systems to track incidents and human errors, supports Over-the-Air (OTA) updates for robot firmware, and manages user accounts with a hierarchical structure of arbitrary depth.

SQLAlchemy, Alembic, Pydantic, Celery, AWS(Lambda, S3, Cognito, EKS, ECR, RDS, SQS, IAM, SageMaker, etc.), Scikit-learn, Tensorflow, HTTP, Kafka, GitLab, GitLab CI/CD, Redis, PostgreSQL, Docker, Docker Compose, Terraform, Prometheus, Grafana, Kubernetes(k8s), Pytest.

Project roles

Software Engineer

Period

05.2019 - 09.2020

Responsibilities

- Backend development;
- Integrating authentication and authorization workflows using Azure Entra ID;
- Implementing background calculations with Azure Functions;
- Connected to Donesafe API to fetch incident reports and error data caused by human factors;
- Writing ETL pipeline using Azure Service Bus and Azure Functions;
- Create and implement SQL database structures to meet application requirements;
- Used stateless HTTP requests for periodic data synchronization and API integrations;
- Analyze SQL query performance using query execution plans and optimize as needed;
- Conduct security and penetration testing on the application;
- Integrated Sentry for error tracking and logging to improve system reliability;
- Optimize SQL databases for performance;

RADIO SYSTEM FOR LARGE WAREHOUSES

Highly loaded radio-based system with statistics collection, live monitoring, remote troubleshooting, etc. Such systems are used in large warehouses like Amazon, Auchan, etc.

- Document the API using Swagger;
- Create scripts and files for automating tasks, as well as packaging and deploying applications in containers;
- Inspect and optimize stored procedures and SQL queries;
- Participate in the configuration of CI/CD processes;
- Write technical documentation to aid in the development and maintenance of the application.

Environment

Python, Django, Django Rest Framework, FastAPI, SQLAlchemy, Pydantic, Alembic, Azure(FUNCTIONS, Container Apps, Entra ID, Cosmos DB, Blob Storage, Service Bus, etc.), BLE, HTTP, Donesafe API, Sigma BI, Redis, PostgreSQL, Swagger, Jenkins, OAuth2, Docker, Docker Compose, Sentry, GitLab, Terraform, Linux, Bash.

Project roles

Software Engineer

Period

07.2018 - 05.2019

Responsibilities

- Backend development;
- Implemented a publish/subscribe model to enable real-time communication between IoT devices and the system using MQTT;
- REST API implementation;
- Azure Blob Storage configuration for storing images and other files;

- Azure Data Factory data migration;
- Refactor existing state management;
- Create technical and API documentation that meets required standards;
- Integrate the application with third-party services;
- Monitor SQL server performance using tools and scripts;
- Creating and maintaining disaster recovery plans for SQL databases;
- Develop serverless part on Azure Functions;
- Create and manage indexes to improve the performance of SQL queries;
- Write automated tests to ensure quality and prevent regressions.

Environment

Python, Django, Django Rest Framework, Azure(Key Vault, Data Factory, SQL, Blob Storage, Functions, App Service, etc.), MQTT, PostgreSQL, RabbitMQ, Celery, Github, Docker, Docker Compose, Pytest, Nginx, Redis.

Professional skills

SKILLS		EXPERIENCE IN YEARS	LAST USED
Programming languages	Python	7	2025
Backend	Django	7	2025
	Django Rest Framework	7	2025
	FastAPI	6	2025
	OAuth2	4	2025
	Celery	6	2025
	SQLAlchemy	6	2025
	Pydantic	6	2025
	Swagger	2	2020
	Alembic	6	2025
	Pytest	6	2025
	Donesafe API	2	2020
Protocols	MQTT	1	2019
	HTTP	4	2022
	BLE	4	2022
Message brokers	RabbitMQ	1	2019
	Kafka	5	2025
	AWS SQS	5	2025

SKILLS		EXPERIENCE IN YEARS	LAST USED
Cloud	Azure Service Bus	2	2020
	Azure	2	2020
	AWS	5	2025
Databases	PostgreSQL	7	2025
	Redis	7	2025
	Cosmos DB	2	2020
	DynamoDB	3	2025
	Timestream	3	2025
Natural Language Processing	Hugging Face	3	2025
	LangChain	3	2025
Machine Learning	Scikit-learn	2	2022
	Tensorflow	2	2022
DevOps	Docker	7	2025
	Docker Compose	7	2025
	Kubernetes(k8s)	5	2025
	Jenkins	4	2025
	GitLab CI/CD	2	2022
	Terraform	6	2025

SKILLS		EXPERIENCE IN YEARS	LAST USED
	Nginx	1	2019
	Grafana	2	2022
	Prometheus	2	2022
	Sentry	2	2020
Source control systems	Git	7	2025
	GitLab	6	2025
	GitHub	1	2019